ADSLM — Adaptive Data Science Language Model

Enterprise AI Evaluation Report

```
======================================================================
ADSLM — ADAPTIVE DATA SCIENCE LANGUAGE MODEL
Enterprise AI Evaluation Report
Generated: 2026-09-04 12:31:29
======================================================================

■■ PIPELINE OVERVIEW
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
Task Type : Clustering
Target Column: None
Best Model : KMeans
Best Score : 0.946

■■ DATASET INSIGHT
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
The dataset contains 164,808 records across 2 features (2 numerical, 0
categorical). The dataset is complete — no missing values detected. Outliers
were detected in 1 column(s) using the IQR method — these were retained for
training but may warrant domain investigation. | Pipeline: median imputation
for continuous features (resistant to skew), mode imputation for categoricals.
IQR-based outlier flagging applied. One-Hot Encoding via pd.get_dummies
(drop_first=True to avoid multicollinearity).

■■ TASK DETECTION
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
No target column was specified. The system applies Unsupervised Learning —
discovering natural groupings in the data without predefined labels. Useful
for operational mode detection and anomaly profiling. | Stratified 80/20
train-test split applied (stratify=y for classification). No data leakage —
preprocessing fit exclusively on training fold.

■■ PREPROCESSING SUMMARY ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
Numerical missing values filled using median imputation. All features
standardised to zero mean and unit variance (StandardScaler). Final feature
matrix contains 2 columns after encoding. | StandardScaler fitted on X_train
only; transform applied to both train/test to prevent test-set leakage.
Boolean dummy columns cast to int8 for memory efficiency.

■■ MODEL RECOMMENDATIONS ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
[1] KMeans
→ Fast, scalable centroid-based clustering. Ideal for partitioning machine
operational modes (e.g., normal, degraded, critical) from unlabeled sensor
telemetry.
[2] DBSCAN
→ Density-based algorithm that discovers clusters of arbitrary shape and
automatically flags outliers as noise. Perfect for anomaly detection in
```

manufacturing process data where defects are rare but critical.

■■ PERFORMANCE METRICS ■■

Silhouette Score : 0.946
N Clusters : 3
N Noise Points : 0

■■ AI MODEL INSIGHT

■■■

After training and comparing all recommended models, KMeans achieved the best performance. It formed 3 coherent clusters with a Silhouette Score of 0.9460 (scores near +1 indicate well-separated, compact clusters). | Model selection criterion: weighted F1 (classification) or RMSE (regression). All models trained with default hyperparameters; recommend GridSearchCV / Optuna for production hyperparameter optimisation.

■■ FEATURE INSIGHT

■■

Feature importance scores are not directly extractable for this model type. Consider using SHAP for model-agnostic explanations. | Importances sourced from estimator's native feature_importances_ (tree-based) or mean(|coef_|) (linear). For model-agnostic explanations, integrate SHAP TreeExplainer / KernelExplainer.

■■ BIG DATA TELEMETRY & SCALE ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

Throughput Speed : 48.5 MB/s (164,808 records)
PySpark RAM Req. : 512.0 MB (2 executors)
Partition Column : 'job_id' (4 partitions)
Storage Format : Delta Lake / Apache Parquet (ACID compliant)

■■ REGULATORY & AI GOVERNANCE AUDIT ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

Compliance Rating: 75 / 100 (ISO 27001 Aligned)
EU AI Act Risk : Minimal Risk (Unsupervised Data Exploratory Analytics)
GDPR Privacy : Clean — No raw PII stored
Human Oversight : Mandatory (Human operator must approve critical fault actions)
→ Voluntary code of conduct and standard documentation.

■■ ACTIONABLE RECOMMENDATIONS ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

1. Investigate extreme sensor spikes detected in columns ['job_id', 'industry_id'] via IQR bounds.
2. Evaluate cluster cohesion (Silhouette Score: 0.9460) using PCA 2D projections.
3. Partition Delta Lake tables on column 'job_id' for sub-second distributed telemetry queries.
4. Deploy the saved model binary via the FastAPI /orchestrate endpoint for production inference.

==

End of ADSLM Report | Enterprise AI Evaluation

==